

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **10**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 10

x -intercept: _____

y -intercept: _____

Find the x -intercept and y -intercept of the line.

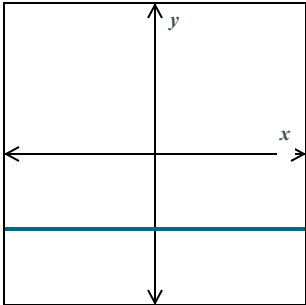
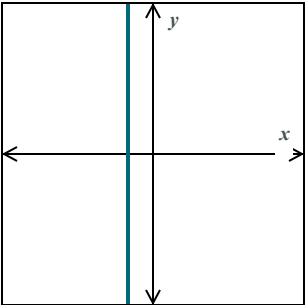
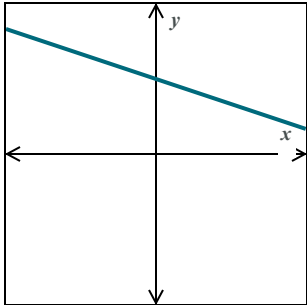
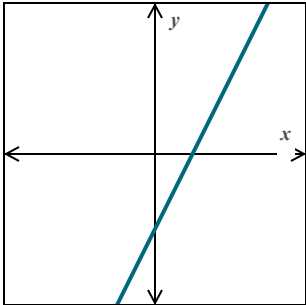
$5x - 4y = -6$

x -intercept: _____

y -intercept: _____

Question 2 of 10

For each line, determine whether the slope is positive, negative, zero, or undefined.

Line 1	Line 2	Line 3	Line 4
			
☐ Positive ☐ Negative ☐ Zero ☐ Undefined	☐ Positive ☐ Negative ☐ Zero ☐ Undefined	☐ Positive ☐ Negative ☐ Zero ☐ Undefined	☐ Positive ☐ Negative ☐ Zero ☐ Undefined

Question 3 of 10

Find the slope of the line passing through the points $(-4, 7)$ and $(2, -8)$.

Question 4 of 10

Find the y -intercept and the slope of the line.

$$-6x + 2y = 4$$

Write your answers in simplest form.

Find the y -intercept and the slope of the line.

$$-6x + 2y = 4$$

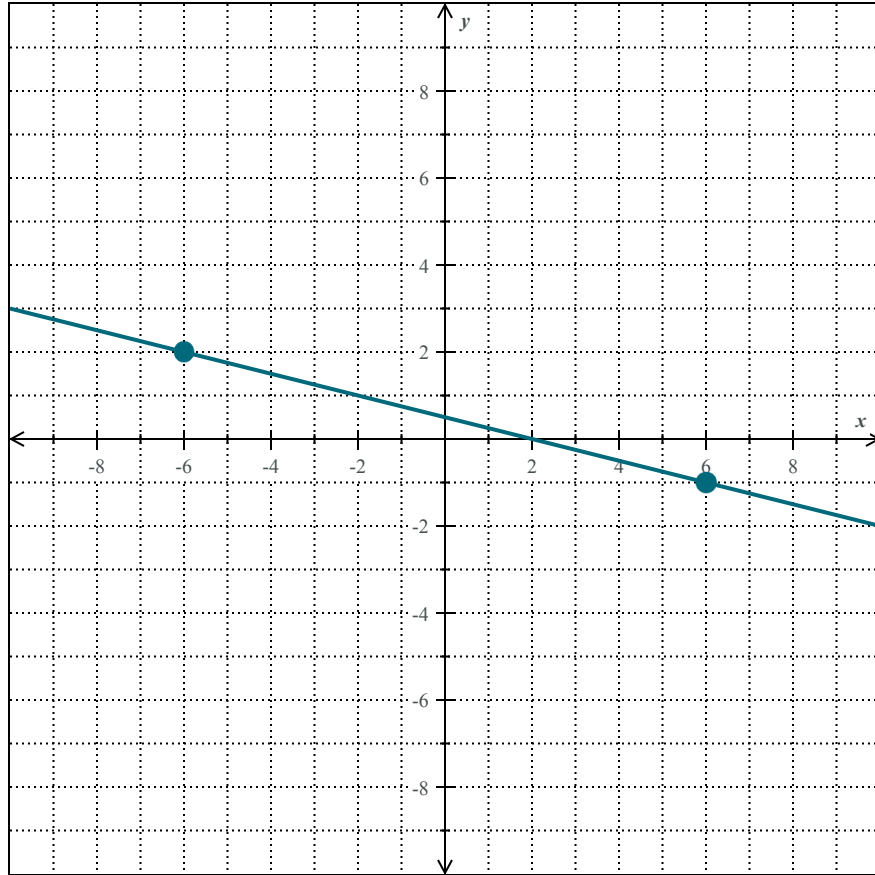
Write your answers in simplest form.

y -intercept: _____

slope: _____

Question 5 of 10

Find an equation for the line below.



Question 6 of 10

Write equations for the vertical and horizontal lines passing through the point $(5, 0)$.

Vertical line:

Horizontal line:

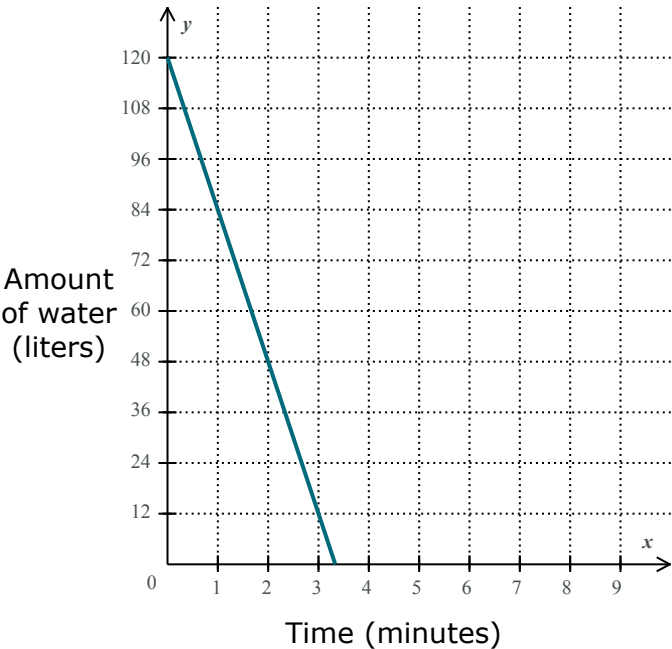
Write equations for the vertical and horizontal lines passing through the point $(5, 0)$.

Vertical line:

Horizontal line:

Question 7 of 10

Deandre is draining an aquarium. The graph shows the amount of water (in liters) in the aquarium versus time (in minutes).



Answer the following questions.

(b) Choose the statement that best describes how the time and amount of water are related. Then give the value requested.

- ☐ As time increases, the amount of water in the aquarium decreases.

At what rate is the amount of water decreasing?

_____ liters per minute

- ☐ As time increases, the amount of water in the aquarium increases.

At what rate is the amount of water increasing?

_____ liters per minute

(b) What is the amount of water in the aquarium at 0 minutes?

_____ liters

Question 8 of 10

Hans is cooking a roast. The table below gives the temperature $R(t)$ of the roast (in degrees Celsius), at a few times t (in minutes) after he removed it from the oven.

Time t (minutes)	Temperature $R(t)$ (°C)
0	226.6
10	205.6
30	157.6
50	119.6
70	61.6

(a) Find the average rate of change for the temperature from 0 minutes to 10 minutes.

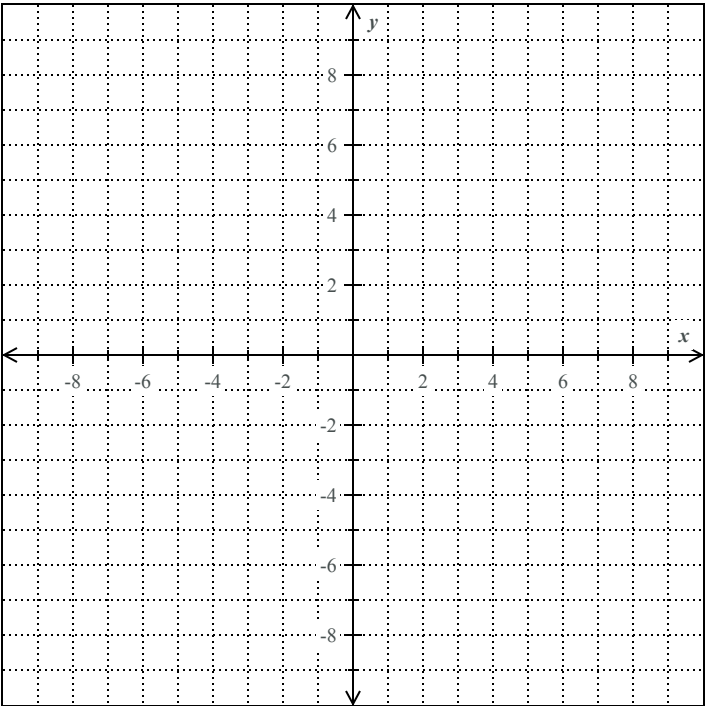
_____ °C per minute

(b) Find the average rate of change for the temperature from 30 minutes to 70 minutes.

_____ °C per minute

Question 9 of 10

Graph the line $y = -1$.



Question 10 of 10

Consider the line $y = -\frac{2}{9}x - 8$.

(a) Find the equation of the line that is perpendicular to this line and passes through the point $(-9, 2)$.

(b) Find the equation of the line that is parallel to this line and passes through the point $(-9, 2)$.

Note that a graphing calculator may be helpful in checking your answer.

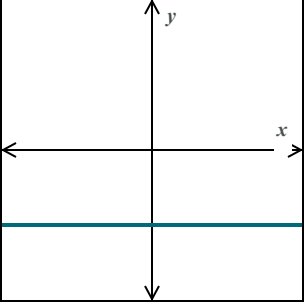
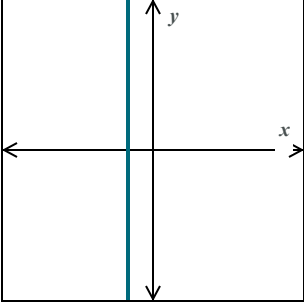
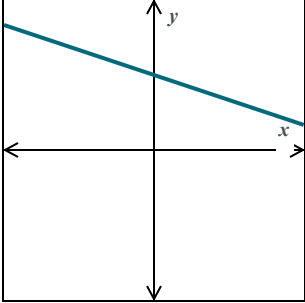
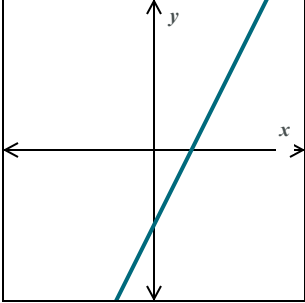
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Question 1 of 10

x -intercept: $-\frac{6}{5}$

y -intercept: $\frac{3}{2}$

Question 2 of 10

Line 1	Line 2	Line 3	Line 4
			
☐ Positive ☐ Negative ☒ Zero ☐ Undefined	☐ Positive ☐ Negative ☐ Zero ☒ Undefined	☐ Positive ☒ Negative ☐ Zero ☐ Undefined	☒ Positive ☐ Negative ☐ Zero ☐ Undefined

Question 3 of 10

$-\frac{5}{2}$

Question 4 of 10

y -intercept: 2

slope: 3

Question 5 of 10

$$y = -\frac{1}{4}x + \frac{1}{2}$$

Note: Other correct answers include $y + 1 = -\frac{1}{4}(x - 6)$ and $x + 4y = 2$.

Question 6 of 10

Vertical line: $x = 5$

Horizontal line: $y = 0$

Question 7 of 10

(a) Choose the statement that best describes how the time and amount of water are related. Then give the value requested.

- ☒ As time increases, the amount of water in the aquarium decreases.

At what rate is the amount of water decreasing?

36 liters per minute

- ☐ As time increases, the amount of water in the aquarium increases.

At what rate is the amount of water increasing?

liters per minute

(b) What is the amount of water in the aquarium at 0 minutes?

120 liters

Question 8 of 10

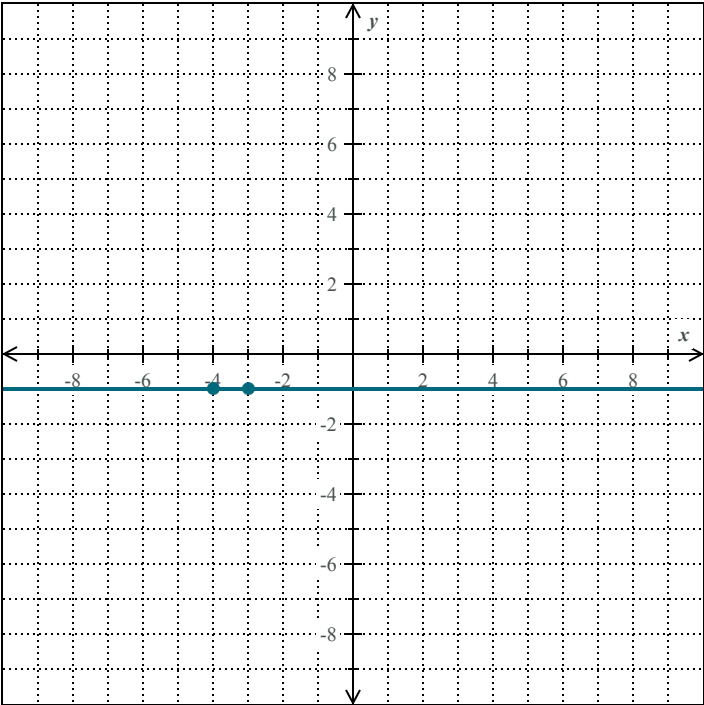
(a) Find the average rate of change for the temperature from 0 minutes to 10 minutes.

−2.1 °C per minute

(b) Find the average rate of change for the temperature from 30 minutes to 70 minutes.

−2.4 °C per minute

Question 9 of 10



Question 10 of 10

Equation of perpendicular line: $y = \frac{9}{2}x + \frac{85}{2}$

Equation of parallel line: $y = -\frac{2}{9}x$